IILM UNIVERSITY, GREATER NOIDA
SCHOOL OF COMPUTER SCIENCE & ENGINEERING
WEEKLY PROGRESS REPORT
Even semester 2026-2027

B.Tech. Major Project Details

Project Id	26P_3CSE024
Project Title	AI-Based Precision Agriculture and Crop Health Monitoring System
Reporting period	30/09/2026 to 06/10/2026
Week No.	1

Work Completed This Week (Phase I: Project Initiation):

- **Problem Definition:** Crop diseases and nutrient or water stress are detected late due to manual, subjective inspection, causing yield loss and excess input use.
- **Literature Survey:** Reviewed CNN-based disease classification (e.g., PlantVillage), NDVI remote sensing and IoT soil sensing; gap: poor field generalisation and few image-plus-sensor systems.
- **Objectives:** To prepare crop image and sensor datasets, design a deep-learning approach for disease and health identification, integrate sensor data, and provide a dashboard with alerts.
- **Scope:** Selected crops and diseases, sensor integration and a web dashboard are included; autonomous machinery control and large-scale commercial deployment are excluded.
- **Novelty:** Combines leaf-image analysis with soil and weather sensor data in one lightweight system giving actionable advisories, unlike image-only solutions.
- **Feasibility Study and Analysis:** Technically feasible with open datasets, Python frameworks and low-cost sensors; economically viable using open-source tools; operationally suitable through a simple dashboard.
- **Multidisciplinary and Transdisciplinary Aspects:** Computer Science (machine learning, computer vision), Agriculture (crop health), Electronics/IoT (sensing) and Data Science (analytics).
- **Product Development Scope:** Prototype web application with disease detection and sensor integration modules, plus project documentation and final report.
- **Identification of Components:** Software: Python, TensorFlow/PyTorch, OpenCV, Flask, database. Hardware: camera, ESP32/Arduino, soil moisture, temperature and humidity sensors, GPU/cloud GPU access.

Challenges/Issues:

- Difficulties faced in problem definition, literature survey or scope finalisation: narrowing a broad literature to selected crops.
- Constraints noted during feasibility analysis or component identification: limited labelled field images and GPU access.
- Action taken/proposed to resolve the above: limit scope to selected crops, add locally collected images, and use cloud GPUs.

Plan for Next Week:

- Finalise the project methodology and approach.
- Prepare the project plan and schedule, and the Work Breakdown Structure (WBS).
- Develop the system design and system architecture.
- Identify project risks and mitigation measures.
- Review progress with the Project Guide and incorporate feedback.

Guide's Remarks:

Signature of Student

Signature of Guide

IILM UNIVERSITY, GREATER NOIDA
SCHOOL OF COMPUTER SCIENCE & ENGINEERING
WEEKLY PROGRESS REPORT
Even semester 2026-2027

B.Tech. Major Project Details

Project Id	26P_3CSE024
Project Title	AI-Based Precision Agriculture and Crop Health Monitoring System
Reporting period	07/10/2026 to 13/10/2026
Week No.	2

Work Completed This Week (Phase II: Planning & Design):

- **Methodology:** Iterative cycle of data preprocessing, CNN/transfer-learning model design (candidates to be compared), sensor integration and dashboard development; evaluation planned via accuracy, precision, recall and F1-score.
- **Project Planning:** Planned phases (initiation, design, implementation, testing, documentation), milestones and module-wise responsibilities.
- **System Design:** Designed modules for image acquisition, preprocessing, disease detection, sensor data, advisory and dashboard, linked via REST APIs.
- **Work Breakdown Structure (WBS):** Decomposed into six work packages: data preparation, model development, sensor integration, backend and dashboard, testing, and documentation.
- **Progress Review:** Reviewed Phase I outcomes (objectives, scope, feasibility, components) as the baseline for planning and design.
- **Project Planning and Scheduling:** Prepared a week-wise timeline with task durations and a milestone at the end of each phase.
- **System Architecture/Design:** Layered architecture: data acquisition (camera, sensors), AI processing, backend, and presentation (dashboard, alerts).
- **Risk Identification:** Risks: limited or imbalanced data (augmentation), overfitting (validation), sensor failure (calibration) and schedule slippage (buffer time).
- **Progress with the Project Guide:** Progress and plan prepared for discussion with the Project Guide; feedback will be recorded in the next weekly report.

Challenges/Issues:

- Difficulties faced in methodology selection, planning or system design: choosing a model suited to limited data; deciding the image–sensor fusion approach.
- Risks or dependencies affecting the schedule: availability of datasets, sensor hardware and GPU resources.
- Action taken/proposed to resolve the above: shortlist candidate models, finalise dataset sources early and add buffer time to the schedule.

Plan for Next Week:

- Incorporate the Project Guide’s feedback into the project plan and system design.
- Commence the next project phase (Implementation) as per the approved schedule: dataset collection, preprocessing and environment setup.
- Review progress with the Project Guide.

Guide's Remarks:

Signature of Student Signature of Guide